

19.8 Principal Angles

The sections so far measured how far eigenvalues move. The next section asks how far **eigenvectors** move, and more generally invariant subspaces, and before it can answer it needs a way to say how far apart two subspaces are. For two lines there is an obvious answer, the angle between them. This section shows that for subspaces of dimension $k \geq 2$ one angle is not enough, and that the right replacement is a list of k angles read off from a singular value decomposition. It then proves the two facts Section 9 will use: the sines of these angles are the singular values of a matrix built from the orthogonal complement, and the largest sine is the distance between the two orthogonal projections.

Throughout, F is $\mathbb{R}$ or $\mathbb{C}$, and F^n carries its standard inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{x}$, with $\mathbf{x}^* = \mathbf{x}^\top$ when $F = \mathbb{R}$. For a subspace U of F^n , P_U is the orthogonal projection onto U (Definition 10.3.2), identified with its matrix in the standard basis. Singular values of a matrix with k columns follow the convention of Chapter 16 §09: $\sigma_i := 0$ for $\min(\text{rows}, k) < i \leq k$, so that σ_i^2 is the i -th largest eigenvalue of $\mathbf{A}^* \mathbf{A}$ for every $i \leq k$ (Eq. 16.9.1).

19.8.1 One angle is not enough

For two lines $\text{span}(\mathbf{u})$ and $\text{span}(\mathbf{w})$ there is one number to compute, the angle between them. For planes the first idea is to copy it: call the angle between U and W the **smallest** angle between a line in U and a line in W .

That idea fails at once in $\mathbb{R}^3$. Two distinct planes $U, W \subseteq \mathbb{R}^3$ always meet in a line, since

$$\dim(U \cap W) = 2 + 2 - \dim(U + W) \geq 4 - 3 = 1$$

by Theorem 1.7.3. So the smallest angle is 0 for **every** pair of planes: it cannot tell a plane tilted by one degree from a plane at right angles.

Taking the **largest** such angle instead does no better in $\mathbb{R}^4$. Let $U = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ and compare

$$W_1 = \text{span}(\mathbf{e}_1, \mathbf{e}_3), \quad W_2 = \text{span}(\mathbf{e}_3, \mathbf{e}_4).$$

Each contains a vector perpendicular to all of U (namely $\mathbf{e}_3$), so each has a line at angle $\pi/2$ from U . But W_1 shares the whole line $\text{span}(\mathbf{e}_1)$ with U , while W_2 is perpendicular to U in every direction. One number cannot record both facts. A list can: $(0, \pi/2)$ for W_1 and $(\pi/2, \pi/2)$ for W_2 . The rest of this section explains where such a list comes from in general, when the subspaces are not lined up with the coordinate axes.

Two subspaces can be given orthonormal bases that face each other in pairs; the principal angles are the angles between the partners.

19.8.2 Principal angles

Everything will be computed from orthonormal bases written as the columns of a matrix, so we first record how such matrices behave.

Let $U \subseteq F^n$ be a subspace with $\dim U = k \geq 1$. An **orthonormal basis matrix** of U is a matrix $\mathbf{Q} \in M_{n \times k}(F)$ whose columns form an orthonormal basis of U . One exists: apply Gram–Schmidt (Theorem 10.2.2) to any basis of U .

Lemma 19.8.1 — Orthonormal Basis Matrices

Let $U \subseteq F^n$ have dimension $k \geq 1$ and let $\mathbf{Q} \in M_{n \times k}(F)$ be an orthonormal basis matrix of U .

- (a) $\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_k$ and $P_U = \mathbf{Q} \mathbf{Q}^*$.
 (b) $\|\mathbf{Q} \mathbf{y}\| = \|\mathbf{y}\|$ for every $\mathbf{y} \in F^k$. For every matrix $\mathbf{Y}$ with k rows and every matrix $\mathbf{Z}$ with k columns,

$$\begin{aligned} \|\mathbf{Q} \mathbf{Y}\|_2 &= \|\mathbf{Y}\|_2, & \|\mathbf{Q} \mathbf{Y}\|_F &= \|\mathbf{Y}\|_F, \\ \|\mathbf{Z} \mathbf{Q}^*\|_2 &= \|\mathbf{Z}\|_2, & \|\mathbf{Z} \mathbf{Q}^*\|_F &= \|\mathbf{Z}\|_F. \end{aligned}$$

- (c) Every orthonormal basis matrix of U is $\mathbf{Q} \mathbf{R}$ for some $\mathbf{R} \in \mathbf{U}(k)$ (orthogonal when $F = \mathbb{R}$).
 (d) $\mathbf{I}_n - P_U = P_{U^\perp}$. If moreover $k < n$ and $\mathbf{Q}_\perp \in M_{n \times (n-k)}(F)$ is an orthonormal basis matrix of $U^\perp$, then the square matrix $(\mathbf{Q} \mid \mathbf{Q}_\perp)$ is unitary, and

$$\mathbf{Q} \mathbf{Q}^* + \mathbf{Q}_\perp \mathbf{Q}_\perp^* = \mathbf{I}_n.$$

Proof.

- (a) The (i, j) entry of $\mathbf{Q}^* \mathbf{Q}$ is $\mathbf{q}_i^* \mathbf{q}_j = \langle \mathbf{q}_j, \mathbf{q}_i \rangle = \delta_{ij}$, because the columns $\mathbf{q}_1, \dots, \mathbf{q}_k$ are orthonormal. By Theorem 10.3.3 (a), for every $\mathbf{v} \in F^n$,

$$P_U \mathbf{v} = \sum_{i=1}^k \langle \mathbf{v}, \mathbf{q}_i \rangle \mathbf{q}_i = \sum_{i=1}^k \mathbf{q}_i (\mathbf{q}_i^* \mathbf{v}) = \mathbf{Q} \mathbf{Q}^* \mathbf{v},$$

the last step being column-times-row expansion of $\mathbf{Q} \mathbf{Q}^*$.

- (b) By (a), $\|\mathbf{Q} \mathbf{y}\|^2 = \mathbf{y}^* \mathbf{Q}^* \mathbf{Q} \mathbf{y} = \mathbf{y}^* \mathbf{y} = \|\mathbf{y}\|^2$. Hence $\|\mathbf{Q} \mathbf{Y} \mathbf{x}\| = \|\mathbf{Y} \mathbf{x}\|$ for every $\mathbf{x}$, and taking the maximum over unit $\mathbf{x}$ (Definition 15.3.2) gives $\|\mathbf{Q} \mathbf{Y}\|_2 = \|\mathbf{Y}\|_2$. For the Frobenius norm, $\|\mathbf{Q} \mathbf{Y}\|_F^2 = \text{tr}(\mathbf{Y}^* \mathbf{Q}^* \mathbf{Q} \mathbf{Y}) = \text{tr}(\mathbf{Y}^* \mathbf{Y}) = \|\mathbf{Y}\|_F^2$. Next, $\|\mathbf{Z} \mathbf{Q}^*\|_F^2 = \text{tr}(\mathbf{Q} \mathbf{Z}^* \mathbf{Z} \mathbf{Q}^*) = \text{tr}(\mathbf{Z}^* \mathbf{Z} \mathbf{Q}^* \mathbf{Q}) = \|\mathbf{Z}\|_F^2$, by Theorem 0.9.6 (c). For the spectral norm, let $\mathbf{x} \in F^n$. By the first sentence and (a), $\|\mathbf{Q}^* \mathbf{x}\| = \|\mathbf{Q} \mathbf{Q}^* \mathbf{x}\| = \|P_U \mathbf{x}\| \leq \|\mathbf{x}\|$ (Theorem 10.3.3 (c)). So $\|\mathbf{Z} \mathbf{Q}^* \mathbf{x}\| \leq \|\mathbf{Z}\|_2 \|\mathbf{Q}^* \mathbf{x}\| \leq \|\mathbf{Z}\|_2 \|\mathbf{x}\|$ by Theorem 15.3.2 (a), and $\|\mathbf{Z} \mathbf{Q}^*\|_2 \leq \|\mathbf{Z}\|_2$. Conversely, choose a unit $\mathbf{y} \in F^k$ with $\|\mathbf{Z} \mathbf{y}\| = \|\mathbf{Z}\|_2$ (Lemma 15.3.1). Then $\mathbf{x} = \mathbf{Q} \mathbf{y}$ is a unit vector with $\mathbf{Z} \mathbf{Q}^* \mathbf{x} = \mathbf{Z} \mathbf{y}$, so $\|\mathbf{Z} \mathbf{Q}^*\|_2 \geq \|\mathbf{Z}\|_2$.
 (c) Let $\mathbf{Q}'$ be another orthonormal basis matrix of U , and put $\mathbf{R} = \mathbf{Q}^* \mathbf{Q}' \in M_k(F)$. Each column of $\mathbf{Q}'$ lies in U , where P_U is the identity, so $\mathbf{Q}' = P_U \mathbf{Q}' = \mathbf{Q} \mathbf{Q}^* \mathbf{Q}' = \mathbf{Q} \mathbf{R}$ by (a). Then $\mathbf{R}^* \mathbf{R} = \mathbf{Q}'^* \mathbf{Q} \mathbf{Q}^* \mathbf{Q}' = \mathbf{Q}'^* P_U \mathbf{Q}' = \mathbf{Q}'^* \mathbf{Q}' = \mathbf{I}_k$, so the square matrix $\mathbf{R}$ is unitary (Definition 10.7.2).
 (d) For the first identity, let $\mathbf{v} \in F^n$ and write $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$, which is possible in exactly one way by Theorem 10.3.2 (a). Then $P_U \mathbf{v} = \mathbf{u}$ by Definition 10.3.2. Since $(U^\perp)^\perp = U$ by Theorem 10.3.2 (b), the splitting $\mathbf{v} = \mathbf{w} + \mathbf{u}$ is also the one that defines $P_{U^\perp}$, so $P_{U^\perp} \mathbf{v} = \mathbf{w} = \mathbf{v} - P_U \mathbf{v}$. As $\mathbf{v}$ was arbitrary, $\mathbf{I}_n - P_U = P_{U^\perp}$. Nothing here restricts k : when $k = n$, $U^\perp = \{\mathbf{0}\}$ and both sides are $\mathbf{0}$.

Now let $k < n$. The columns of $\mathbf{Q}$ and of $\mathbf{Q}_\perp$ are orthonormal, and each column of $\mathbf{Q}$ is orthogonal to each column of $\mathbf{Q}_\perp$ because $U \perp U^\perp$. Their number is $k + (n - k) = n$, by

Theorem 10.3.2 (c). So the square matrix $\mathbf{O} = (\mathbf{Q} \mid \mathbf{Q}_\perp) \in M_n(F)$ satisfies $\mathbf{O}^* \mathbf{O} = \mathbf{I}_n$, hence is unitary, and then $\mathbf{O} \mathbf{O}^* = \mathbf{I}_n$ as well, since $\mathbf{O}^* = \mathbf{O}^{-1}$. By block multiplication (Theorem 7.1.1), $\mathbf{O} \mathbf{O}^* = \mathbf{Q} \mathbf{Q}^* + \mathbf{Q}_\perp \mathbf{Q}_\perp^*$. This also re-proves the first identity here, since $\mathbf{Q}_\perp \mathbf{Q}_\perp^* = P_{U^\perp}$ by (a) applied to $U^\perp$.

■

Now the definition. The question is how much of U survives when it is projected into W . By parts (a) and (b) of the lemma, for a unit vector $\mathbf{u} = \mathbf{Q}_U \mathbf{y}$ of U the surviving length is $\|P_W \mathbf{u}\| = \|\mathbf{Q}_W \mathbf{Q}_W^* \mathbf{u}\| = \|\mathbf{Q}_W^* \mathbf{u}\| = \|(\mathbf{Q}_W^* \mathbf{Q}_U) \mathbf{y}\|$. So the matrix $\mathbf{Q}_W^* \mathbf{Q}_U$ records the answer for every direction of U at once, and its singular values are the stretch factors along the best directions.

Definition 19.8.1 — Principal Angles

Let $U, W \subseteq F^n$ be subspaces with

$$1 \leq k = \dim U \leq l = \dim W,$$

and let $\mathbf{Q}_U \in M_{n \times k}(F)$ and $\mathbf{Q}_W \in M_{n \times l}(F)$ be orthonormal basis matrices of U and W . The **principal angles** between U and W are the k numbers

$$0 \leq \theta_1 \leq \theta_2 \leq \cdots \leq \theta_k \leq \frac{\pi}{2}$$

determined by

$$\cos \theta_i = \sigma_i(\mathbf{Q}_W^* \mathbf{Q}_U) \quad (i = 1, \dots, k).$$

We write $\Theta(U, W) = \text{diag}(\theta_1, \dots, \theta_k)$ and $\sin \Theta(U, W) = \text{diag}(\sin \theta_1, \dots, \sin \theta_k)$.

In words: fix orthonormal bases of the two subspaces, form the $l \times k$ matrix whose (i, j) entry is the inner product of the j -th basis vector of U with the i -th basis vector of W , and take its k singular values. These lie in $[0, 1]$, as we check next, so each is the cosine of exactly one angle in $[0, \pi/2]$. The largest cosine gives the **smallest** angle, so the decreasing singular values produce increasing angles. The number of angles is the **smaller** dimension, k , and the definition asks for $k \leq l$; for a pair with $\dim U > \dim W$, swap the names. The two notations at the end are for use in norms: $\|\sin \Theta(U, W)\|_2 = \sin \theta_k$, the largest entry of a non-negative diagonal matrix (Lemma 16.3.5), and $\|\sin \Theta(U, W)\|_F = (\sum_i \sin^2 \theta_i)^{1/2}$.

Well-definedness. Three things need checking. The answer must not depend on the orthonormal bases chosen. The singular values must be at most 1, or there is no angle. And when $k = l$ the definition treats U and W asymmetrically, so we should check that swapping them changes nothing.

Proposition 19.8.2 — Principal Angles Are Well Defined

In the setting of Definition 19.8.1, put $\mathbf{M} = \mathbf{Q}_W^* \mathbf{Q}_U \in M_{l \times k}(F)$.

- (a) The singular values of $\mathbf{M}$ do not depend on the choice of the orthonormal basis matrices $\mathbf{Q}_U$ and $\mathbf{Q}_W$.
- (b) $1 \geq \sigma_1(\mathbf{M}) \geq \cdots \geq \sigma_k(\mathbf{M}) \geq 0$, so $\theta_1, \dots, \theta_k$ exist, are unique, and are non-decreasing.

- (c) $\sigma_i(\mathbf{Q}_U^* \mathbf{Q}_W) = \sigma_i(\mathbf{M})$ for $i = 1, \dots, k$. In particular, if $k = l$, the principal angles between W and U are those between U and W .
- (d) If $k = 1$ and $U = \text{span}(\mathbf{u})$ with $\|\mathbf{u}\| = 1$, then $\cos \theta_1 = \|P_W \mathbf{u}\|$. If also $l = 1$ and $W = \text{span}(\mathbf{w})$ with $\|\mathbf{w}\| = 1$, then $\cos \theta_1 = |\langle \mathbf{u}, \mathbf{w} \rangle|$.

Proof.

- (a) By Lemma 19.8.1 (c), any other choices are $\mathbf{Q}_U \mathbf{R}$ and $\mathbf{Q}_W \mathbf{R}'$ with $\mathbf{R} \in U(k)$ and $\mathbf{R}' \in U(l)$. The new matrix is $\mathbf{M}' = \mathbf{R}'^* \mathbf{M} \mathbf{R}$, and

$$\mathbf{M}'^* \mathbf{M}' = \mathbf{R}^* \mathbf{M}^* \mathbf{R}'^* \mathbf{R}' \mathbf{M} \mathbf{R} = \mathbf{R}^* (\mathbf{M}^* \mathbf{M}) \mathbf{R},$$

using $\mathbf{R}' \mathbf{R}'^* = \mathbf{I}_l$. Since $\mathbf{R}^* = \mathbf{R}^{-1}$, this is similar to $\mathbf{M}^* \mathbf{M}$, so the two have the same characteristic polynomial (Theorem 6.10.3) and the same eigenvalues with multiplicity. By Definition 12.8.1, $\mathbf{M}'$ and $\mathbf{M}$ have the same singular values.

- (b) By Theorem 15.3.2 (d) and Lemma 19.8.1 (b), $\|\mathbf{M}\|_2 \leq \|\mathbf{Q}_W^*\|_2 \|\mathbf{Q}_U\|_2$, and $\|\mathbf{Q}_U\|_2 = \|\mathbf{Q}_U \mathbf{I}_k\|_2 = \|\mathbf{I}_k\|_2 = 1$ by Theorem 15.3.2 (c). Also $\|\mathbf{Q}_W^* \mathbf{x}\| = \|P_W \mathbf{x}\| \leq \|\mathbf{x}\|$, as in the proof of that lemma, so $\|\mathbf{Q}_W^*\|_2 \leq 1$. Hence $\sigma_1(\mathbf{M}) = \|\mathbf{M}\|_2 \leq 1$ by Theorem 15.3.3 (c), and all singular values lie in $[0, 1]$. The cosine is a strictly decreasing bijection from $[0, \pi/2]$ onto $[0, 1]$, so each $\sigma_i(\mathbf{M})$ is $\cos \theta_i$ for exactly one $\theta_i \in [0, \pi/2]$, and $\sigma_1 \geq \dots \geq \sigma_k$ gives $\theta_1 \leq \dots \leq \theta_k$.
- (c) Let $\mathbf{M} = \mathbf{Y} \mathbf{\Sigma} \mathbf{Z}^*$ be a singular value decomposition (Theorem 12.8.1). Then $\mathbf{M}^* = \mathbf{Z} \mathbf{\Sigma}^T \mathbf{Y}^*$, where $\mathbf{Z}, \mathbf{Y}$ are unitary and $\mathbf{\Sigma}^T \in M_{k \times l}(\mathbb{R})$ carries the same diagonal entries $\sigma_1(\mathbf{M}) \geq \dots \geq \sigma_k(\mathbf{M})$ and zeros elsewhere. So this is a factorization of $\mathbf{M}^*$ of the shape in Theorem 12.8.1, and Theorem 12.8.2 gives $\sigma_i(\mathbf{M}^*) = \sigma_i(\mathbf{M})$. Since $\mathbf{M}^* = \mathbf{Q}_U^* \mathbf{Q}_W$, this is the first claim. When $k = l$, the principal angles between W and U are defined by $\sigma_i(\mathbf{Q}_U^* \mathbf{Q}_W)$, so they coincide with those between U and W .
- (d) With $\mathbf{Q}_U = \mathbf{u}$, the matrix $\mathbf{M} = \mathbf{Q}_W^* \mathbf{u}$ is a single column, and its one singular value is $(\mathbf{M}^* \mathbf{M})^{1/2} = \|\mathbf{Q}_W^* \mathbf{u}\|$. By Lemma 19.8.1 (b) and (a), $\|\mathbf{Q}_W^* \mathbf{u}\| = \|\mathbf{Q}_W \mathbf{Q}_W^* \mathbf{u}\| = \|P_W \mathbf{u}\|$. If $l = 1$ and $\mathbf{Q}_W = \mathbf{w}$, then $\mathbf{M} = \mathbf{w}^* \mathbf{u} = \langle \mathbf{u}, \mathbf{w} \rangle$, whose singular value is $|\langle \mathbf{u}, \mathbf{w} \rangle|$.

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Examples.

- Two lines in $\mathbb{R}^2$.** For $\mathbf{u} = \mathbf{e}_1$ and $\mathbf{w} = (1, 1)/\sqrt{2}$, part (d) gives $\cos \theta_1 = 1/\sqrt{2}$, so $\theta_1 = \pi/4$. Replacing $\mathbf{w}$ by $-\mathbf{w}$ changes neither the line nor the answer.
- Nested subspaces.** If $U \subseteq W$, then every column of $\mathbf{Q}_U$ is fixed by P_W , so $\mathbf{M}^* \mathbf{M} = \mathbf{Q}_U^* \mathbf{Q}_W \mathbf{Q}_W^* \mathbf{Q}_U = \mathbf{Q}_U^* P_W \mathbf{Q}_U = \mathbf{Q}_U^* \mathbf{Q}_U = \mathbf{I}_k$. All singular values are 1 and all principal angles are 0. The degenerate case $U = W$ is included, and it is the reason the angles start at 0.
- Perpendicular subspaces.** If $U \subseteq W^\perp$, every entry of $\mathbf{M}$ is the inner product of a vector of U with a vector of W , hence 0, so all principal angles are $\pi/2$.
- The planes of the opening.** For $U = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ and $W_1 = \text{span}(\mathbf{e}_1, \mathbf{e}_3)$, take the standard vectors as orthonormal bases. Then $\mathbf{M} = \mathbf{Q}_{W_1}^T \mathbf{Q}_U = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, with singular values 1, 0, so the principal angles are $(0, \pi/2)$. For $W_2 = \text{span}(\mathbf{e}_3, \mathbf{e}_4)$ we are in Example 3, with angles $(\pi/2, \pi/2)$. These are the two lists that the single-number attempts could not tell apart.

Non-example by minimal change. In Example 4, replace the orthonormal basis $(\mathbf{e}_1, \mathbf{e}_3)$ of W_1 by the basis $(\mathbf{e}_1, (\mathbf{e}_1 + \mathbf{e}_3)/\sqrt{2})$, whose vectors still have length 1 but are no longer orthogonal. The matrix of inner products becomes $\begin{pmatrix} 1 & 0 \\ 1/\sqrt{2} & 0 \end{pmatrix}$, with largest singular value $\sqrt{3/2} > 1$, and no angle has that cosine. What fails is the orthogonality of the columns of $\mathbf{Q}_W$: without it, $\mathbf{Q}_W \mathbf{Q}_W^*$ is not the projection P_W , and the matrix no longer measures how much of U survives in W .

Why this definition: the principal vectors. The singular value decomposition of $\mathbf{M}$ does more than produce numbers. It produces bases of U and W in which the whole configuration is visible, and this is the precise meaning of the slogan.

Proposition 19.8.3 — Principal Vectors

In the setting of Definition 19.8.1, let $\mathbf{Q}_W^* \mathbf{Q}_U = \mathbf{Y} \mathbf{\Sigma} \mathbf{Z}^*$ be a singular value decomposition, with $\mathbf{Y} \in M_l(F)$ and $\mathbf{Z} \in M_k(F)$ unitary, columns $\mathbf{y}_1, \dots, \mathbf{y}_l$ and $\mathbf{z}_1, \dots, \mathbf{z}_k$. Put

$$\mathbf{u}_i = \mathbf{Q}_U \mathbf{z}_i \quad (1 \leq i \leq k), \quad \mathbf{w}_j = \mathbf{Q}_W \mathbf{y}_j \quad (1 \leq j \leq l).$$

Then $(\mathbf{u}_1, \dots, \mathbf{u}_k)$ is an orthonormal basis of U , $(\mathbf{w}_1, \dots, \mathbf{w}_l)$ is an orthonormal basis of W , and

$$\langle \mathbf{u}_i, \mathbf{w}_j \rangle = \begin{cases} \cos \theta_i & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Proof. The matrix $\mathbf{Q}_U \mathbf{Z}$ has columns $\mathbf{u}_1, \dots, \mathbf{u}_k$, all in U , and $(\mathbf{Q}_U \mathbf{Z})^* (\mathbf{Q}_U \mathbf{Z}) = \mathbf{Z}^* \mathbf{Z} = \mathbf{I}_k$ by Lemma 19.8.1 (a). So the $\mathbf{u}_i$ are k orthonormal vectors of U . They are linearly independent (Theorem 10.1.4), and $\dim U = k$, so they form a basis. The same argument applies to the $\mathbf{w}_j$ in W . Finally,

$$\langle \mathbf{u}_i, \mathbf{w}_j \rangle = \mathbf{w}_j^* \mathbf{u}_i = \mathbf{y}_j^* (\mathbf{Q}_W^* \mathbf{Q}_U) \mathbf{z}_i = \mathbf{y}_j^* \mathbf{Y} \mathbf{\Sigma} \mathbf{Z}^* \mathbf{z}_i = \mathbf{e}_j^* \mathbf{\Sigma} \mathbf{e}_i,$$

which is the (j, i) entry of $\mathbf{\Sigma}$: it equals $\sigma_i = \cos \theta_i$ when $j = i$, and 0 otherwise. ■

So U and W have orthonormal bases that face each other in pairs. Each $\mathbf{u}_i$ makes the angle θ_i with its partner $\mathbf{w}_i$, in the sense of part (d) of the proposition, and is perpendicular to every other $\mathbf{w}_j$. The last $l - k$ vectors of W have no partner, and they are perpendicular to all of U . The vectors $\mathbf{u}_i, \mathbf{w}_i$ are called **principal vectors**. Two subspaces, however they sit in F^n , look like a list of k independent pairs of lines, and the principal angles are the angles within the pairs.

The principal vectors are usually **not** the basis vectors one started with. For $U = W = \mathbb{R}^2$ with bases $(\mathbf{e}_1, \mathbf{e}_2)$ and $((\mathbf{e}_1 + \mathbf{e}_2)/\sqrt{2}, (\mathbf{e}_1 - \mathbf{e}_2)/\sqrt{2})$, each given basis vector of U is at angle $\pi/4$ from each given basis vector of W . The principal angles are nevertheless 0, 0, by Example 2, since the subspaces are equal. Pairing the given columns one by one measures the bases, not the subspaces.

Warning. **Lines have no orientation, so principal angles use $|\langle \mathbf{u}, \mathbf{w} \rangle|$ and lie in $[0, \pi/2]$.** The angle of Definition 10.1.5 is between **vectors** in a **real** space, and it can reach π . For $\mathbf{u} = \mathbf{e}_1$ and $\mathbf{w} = -\mathbf{e}_1$ in $\mathbb{R}^2$ that angle is π , yet $\text{span}(\mathbf{u}) = \text{span}(\mathbf{w})$ and the principal angle is 0. Over $\mathbb{C}$ the vector angle is not even defined: $\mathbf{u} = \mathbf{e}_1$ and $\mathbf{w} = i\mathbf{e}_1$ span the same complex line, $\langle \mathbf{u}, \mathbf{w} \rangle = i = -i$ is not real, and the principal angle is again

0 because $|-i| = 1$.

Quick check. Find the principal angle between the line $U = \text{span}((1, 2, 2))$ and the plane $W = \{\mathbf{x} \in \mathbb{R}^3 : x_3 = 0\}$.

Solution. Here $k = 1 \leq 2 = l$. The unit vector $\mathbf{u} = \frac{1}{3}(1, 2, 2)$ spans U , and $P_W \mathbf{u} = \frac{1}{3}(1, 2, 0)$, since $(\mathbf{e}_1, \mathbf{e}_2)$ is an orthonormal basis of W (Theorem 10.3.3 (a)). By [Proposition 19.8.2](#) (d), $\cos \theta_1 = \|P_W \mathbf{u}\| = \sqrt{5}/3$. So $\theta_1 = \arccos(\sqrt{5}/3) \approx 41.8^\circ$, and $\sin \theta_1 = 2/3$, which is the distance from $\mathbf{u}$ to the plane.

19.8.3 Sines and the orthogonal complement

The cosines measure how much of U survives projection into W . What Section 9 can actually bound is the part that does **not** survive: the component of U in $W^\perp$. Pythagoras says the two parts are complementary, and the theorem makes this exact.

Theorem 19.8.4 — Sines of Principal Angles

Let $U, W \subseteq F^n$ with $1 \leq k = \dim U \leq l = \dim W$, principal angles $\theta_1 \leq \dots \leq \theta_k$, and orthonormal basis matrices $\mathbf{Q}_U, \mathbf{Q}_W$.

- (a) Suppose $l < n$, and let $\mathbf{Q}_{W^\perp} \in M_{n \times (n-l)}(F)$ be an orthonormal basis matrix of $W^\perp$. Put $\mathbf{M} = \mathbf{Q}_W^* \mathbf{Q}_U$ and $\mathbf{N} = \mathbf{Q}_{W^\perp}^* \mathbf{Q}_U \in M_{(n-l) \times k}(F)$. Then

$$\mathbf{M}^* \mathbf{M} + \mathbf{N}^* \mathbf{N} = \mathbf{I}_k,$$

and the eigenvalues of $\mathbf{N}^* \mathbf{N}$, with multiplicity, are $\sin^2 \theta_1, \dots, \sin^2 \theta_k$. Equivalently, $\sigma_i(\mathbf{N}) = \sin \theta_{k+1-i}$ for $i = 1, \dots, k$, with the convention for σ_i stated at the start of the section.

- (b) For every $l \leq n$,

$$\|(\mathbf{I} - P_W)P_U\|_2 = \|(\mathbf{I} - P_W)\mathbf{Q}_U\|_2 = \sin \theta_k = \|\sin \Theta(U, W)\|_2,$$

and

$$\|(\mathbf{I} - P_W)P_U\|_F = \|(\mathbf{I} - P_W)\mathbf{Q}_U\|_F = \|\sin \Theta(U, W)\|_F.$$

Idea. Split the identity of F^n along $W \oplus W^\perp$ and sandwich it between $\mathbf{Q}_U^*$ and $\mathbf{Q}_U$. That gives two Hermitian $k \times k$ matrices adding up to $\mathbf{I}_k$, and such matrices share their eigenvectors, with eigenvalues c and $1 - c$. The first has eigenvalues $\cos^2 \theta_i$ by definition, so the second has eigenvalues $\sin^2 \theta_i$. Part (b) then removes the orthonormal factors, which change no norm.

Proof.

- (a) By [Lemma 19.8.1](#) (d) applied to W , $\mathbf{I}_n = \mathbf{Q}_W \mathbf{Q}_W^* + \mathbf{Q}_{W^\perp} \mathbf{Q}_{W^\perp}^*$. Multiplying on the left by $\mathbf{Q}_U^*$ and on the right by $\mathbf{Q}_U$, and using $\mathbf{Q}_U^* \mathbf{Q}_U = \mathbf{I}_k$ (part (a) of the lemma), gives $\mathbf{I}_k = \mathbf{M}^* \mathbf{M} + \mathbf{N}^* \mathbf{N}$.

The matrix $\mathbf{M}^* \mathbf{M} \in M_k(F)$ is Hermitian (real symmetric when $F = \mathbb{R}$). By

Corollary 11.4.3 (or Corollary 11.5.2 when $F = \mathbb{R}$) there is an orthonormal basis $\mathbf{z}_1, \dots, \mathbf{z}_k$ of F^k consisting of eigenvectors of $\mathbf{M}^*\mathbf{M}$. Since $\mathbf{M}$ has $l \geq k$ rows, all k eigenvalues of $\mathbf{M}^*\mathbf{M}$ are squared singular values (Definition 12.8.1), so we may number the $\mathbf{z}_i$ so that $\mathbf{M}^*\mathbf{M}\mathbf{z}_i = \sigma_i(\mathbf{M})^2 \mathbf{z}_i = \cos^2 \theta_i \mathbf{z}_i$. Then

$$\mathbf{N}^*\mathbf{N}\mathbf{z}_i = (\mathbf{I}_k - \mathbf{M}^*\mathbf{M})\mathbf{z}_i = (1 - \cos^2 \theta_i)\mathbf{z}_i = \sin^2 \theta_i \mathbf{z}_i.$$

So the unitary matrix with columns $\mathbf{z}_1, \dots, \mathbf{z}_k$ diagonalizes $\mathbf{N}^*\mathbf{N}$ with diagonal $\sin^2 \theta_1, \dots, \sin^2 \theta_k$, and these are its eigenvalues with multiplicity (Theorem 6.10.3). The sine is increasing on $[0, \pi/2]$, so in decreasing order they read $\sin^2 \theta_k \geq \dots \geq \sin^2 \theta_1 \geq 0$. By Eq. 16.9.1, the i -th largest eigenvalue of $\mathbf{N}^*\mathbf{N}$ is $\sigma_i(\mathbf{N})^2$ for every $i \leq k$, with the stated convention. Taking non-negative square roots gives $\sigma_i(\mathbf{N}) = \sin \theta_{k+1-i}$.

- (b) If $l = n$, then $W = F^n$, so $U \subseteq W$ and every $\theta_i = 0$ (Example 2 above), while $\mathbf{I} - P_W = \mathbf{0}$; all four norms are 0. Suppose $l < n$. By Lemma 19.8.1 (d) and (a),

$$(\mathbf{I} - P_W)\mathbf{Q}_U = \mathbf{Q}_{W^\perp}\mathbf{Q}_{W^\perp}^*\mathbf{Q}_U = \mathbf{Q}_{W^\perp}\mathbf{N},$$

and $(\mathbf{I} - P_W)P_U = \mathbf{Q}_{W^\perp}\mathbf{N}\mathbf{Q}_U^*$. By part (b) of the lemma, applied to $\mathbf{Q}_{W^\perp}$ and then to $\mathbf{Q}_U$, both matrices have the spectral and Frobenius norms of $\mathbf{N}$. Now $\|\mathbf{N}\|_2 = \sigma_1(\mathbf{N}) = \sin \theta_k$ by Theorem 15.3.3 (c) and part (a). Also $\|\mathbf{N}\|_F^2 = \text{tr}(\mathbf{N}^*\mathbf{N})$, which is the sum of the eigenvalues of $\mathbf{N}^*\mathbf{N}$ (Theorem 8.3.3), that is $\sum_i \sin^2 \theta_i = \|\sin \Theta(U, W)\|_F^2$. This proves the theorem. ■

Part (b) has a plain geometric reading. For a unit vector $\mathbf{u} \in U$, the vector $(\mathbf{I} - P_W)\mathbf{u}$ is the error in approximating $\mathbf{u}$ from W , and its length is the distance from $\mathbf{u}$ to W (Theorem 10.3.5). So $\sin \theta_k$ is the **largest distance from a unit vector of U to W** , and θ_k is the worst angle any direction of U makes with W . That is the number to bound when one asks whether a computed subspace is close to the true one, and part (a) turns it into the norm of the concrete matrix $\mathbf{Q}_{W^\perp}^*\mathbf{Q}_U$. In Section 9, U will be an invariant subspace of $\mathbf{A}$ and W the corresponding one for $\mathbf{A} + \mathbf{E}$. The columns of $\mathbf{Q}_{W^\perp}$ will be the remaining eigenvectors of $\mathbf{A} + \mathbf{E}$, and $\mathbf{N}$ will turn out to satisfy a Sylvester equation, so the bounds of Section 7 will apply to it.

19.8.4 The distance between two subspaces

A subspace is determined by its orthogonal projection, since $U = \text{im } P_U$ by Theorem 3.10.1 (b), applied to $F^n = U \oplus U^\perp$. So a second natural way to compare U and W is the size of $P_U - P_W$. When the dimensions agree, this measures exactly the largest principal angle.

Theorem 19.8.5 — The Distance Between Projections

Let $U, W \subseteq F^n$ with $\dim U = \dim W = k \geq 1$, and principal angles $\theta_1 \leq \dots \leq \theta_k$. Then

- (a) $\|P_U - P_W\|_2 = \sin \theta_k$;
- (b) $\|P_U - P_W\|_F = \sqrt{2} \|\sin \Theta(U, W)\|_F$;
- (c) $\|P_U - P_W\|_2 \leq 1$, with equality if and only if $U \cap W^\perp \neq \{\mathbf{0}\}$.

Idea. For (b), the **squared** Frobenius norm of the Hermitian matrix $\mathbf{D} = P_U - P_W$ is $\text{tr}(\mathbf{D}^2)$, which expands into traces of projections: $k + k - 2 \text{tr}(P_U P_W)$, and $\text{tr}(P_U P_W) = \sum \cos^2 \theta_i$. For (a), split $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2$ along $U \oplus U^\perp$. Then $\mathbf{D}$ sends $\mathbf{x}_1$ to $(\mathbf{I} - P_W)\mathbf{x}_1 \in W^\perp$ and $\mathbf{x}_2$ to $-P_W \mathbf{x}_2 \in W$, so Pythagoras separates the two contributions. The first is controlled by the sines for (U, W) , the second by the sines for (W, U) , and equal dimensions make these the same list.

Proof. Put $\mathbf{D} = P_U - P_W$, and let $\mathbf{Q}_U, \mathbf{Q}_W$ be orthonormal basis matrices and $\mathbf{M} = \mathbf{Q}_W^* \mathbf{Q}_U \in M_k(F)$.

- (b) By Lemma 19.8.1 (a), $P_U = \mathbf{Q}_U \mathbf{Q}_U^*$ and $P_W = \mathbf{Q}_W \mathbf{Q}_W^*$ are Hermitian and satisfy $P_U^2 = P_U$, $P_W^2 = P_W$. Hence $\mathbf{D}$ is Hermitian and

$$\begin{aligned} \|\mathbf{D}\|_F^2 &= \text{tr}(\mathbf{D}^* \mathbf{D}) = \text{tr}(\mathbf{D}^2) \\ &= \text{tr}(P_U) + \text{tr}(P_W) - \text{tr}(P_U P_W) - \text{tr}(P_W P_U). \end{aligned}$$

By Theorem 0.9.6 (c), $\text{tr}(P_U) = \text{tr}(\mathbf{Q}_U^* \mathbf{Q}_U) = \text{tr}(\mathbf{I}_k) = k$, likewise $\text{tr}(P_W) = k$, and $\text{tr}(P_W P_U) = \text{tr}(P_U P_W)$. Also

$$\begin{aligned} \text{tr}(P_U P_W) &= \text{tr}(\mathbf{Q}_U \mathbf{Q}_U^* \mathbf{Q}_W \mathbf{Q}_W^*) \\ &= \text{tr}(\mathbf{Q}_W^* \mathbf{Q}_U \mathbf{Q}_U^* \mathbf{Q}_W) = \text{tr}(\mathbf{M} \mathbf{M}^*), \end{aligned}$$

and $\text{tr}(\mathbf{M} \mathbf{M}^*) = \|\mathbf{M}^*\|_F^2 = \|\mathbf{M}\|_F^2 = \text{tr}(\mathbf{M}^* \mathbf{M}) = \sum_i \cos^2 \theta_i$, since the eigenvalues of $\mathbf{M}^* \mathbf{M}$ are $\cos^2 \theta_1, \dots, \cos^2 \theta_k$ (Definition 12.8.1, Theorem 8.3.3). Therefore

$$\|\mathbf{D}\|_F^2 = 2k - 2 \sum_{i=1}^k \cos^2 \theta_i = 2 \sum_{i=1}^k \sin^2 \theta_i,$$

which is (b).

- (c) Put $s = \sin \theta_k$. By Theorem 19.8.4 (b), $\|(\mathbf{I} - P_W)P_U\|_2 = s$. The same theorem with the roles of U and W exchanged, which is allowed because $\dim W = \dim U$, gives $\|(\mathbf{I} - P_U)P_W\|_2 = s$ too, because by Proposition 19.8.2 (c) the principal angles between W and U are those between U and W .

Let $\mathbf{x} \in F^n$, and write $\mathbf{x}_1 = P_U \mathbf{x} \in U$ and $\mathbf{x}_2 = \mathbf{x} - \mathbf{x}_1 \in U^\perp$ (Definition 10.3.2). Then $P_U \mathbf{x} = \mathbf{x}_1$ and $P_W \mathbf{x} = P_W \mathbf{x}_1 + P_W \mathbf{x}_2$, so

$$\mathbf{D} \mathbf{x} = (\mathbf{I} - P_W) \mathbf{x}_1 - P_W \mathbf{x}_2.$$

The first term lies in $W^\perp$ and the second in W , by Theorem 10.3.2, so they are orthogonal, and by Theorem 10.1.3

$$\|\mathbf{D} \mathbf{x}\|^2 = \|(\mathbf{I} - P_W) \mathbf{x}_1\|^2 + \|P_W \mathbf{x}_2\|^2. \quad (19.8.1)$$

We bound the two terms. Since $\mathbf{x}_1 = P_U \mathbf{x}_1$, $\|(\mathbf{I} - P_W) \mathbf{x}_1\| = \|(\mathbf{I} - P_W) P_U \mathbf{x}_1\| \leq s \|\mathbf{x}_1\|$ by Theorem 15.3.2 (a). For the second term, P_W is Hermitian and idempotent and $\mathbf{x}_2 = (\mathbf{I} - P_U) \mathbf{x}_2$, so

$$\begin{aligned} \|P_W \mathbf{x}_2\|^2 &= \langle P_W \mathbf{x}_2, \mathbf{x}_2 \rangle = \langle P_W \mathbf{x}_2, (\mathbf{I} - P_U) \mathbf{x}_2 \rangle \\ &= \langle (\mathbf{I} - P_U) P_W (P_W \mathbf{x}_2), \mathbf{x}_2 \rangle \leq s \|P_W \mathbf{x}_2\| \|\mathbf{x}_2\|, \end{aligned}$$

using Theorem 10.3.3 (b) for $\mathbf{I} - P_U = P_{U^\perp}$ (Lemma 19.8.1 (d)), then $P_W = P_W^2$, then Cauchy–Schwarz (Theorem 10.1.5) with $\|(\mathbf{I} - P_U)P_W\|_2 = s$. Hence $\|P_W \mathbf{x}_2\| \leq s\|\mathbf{x}_2\|$, the case $P_W \mathbf{x}_2 = \mathbf{0}$ being trivial. Adding, and using $\|\mathbf{x}_1\|^2 + \|\mathbf{x}_2\|^2 = \|\mathbf{x}\|^2$ (Theorem 10.1.3),

$$\|\mathbf{D}\mathbf{x}\|^2 \leq s^2(\|\mathbf{x}_1\|^2 + \|\mathbf{x}_2\|^2) = s^2\|\mathbf{x}\|^2.$$

So $\|\mathbf{D}\|_2 \leq s$. For the reverse, Theorem 19.8.4 (b) gives $\|(\mathbf{I} - P_W)\mathbf{Q}_U\|_2 = s$, and by Lemma 15.3.1 there is a unit $\mathbf{y} \in F^k$ with $\|(\mathbf{I} - P_W)\mathbf{Q}_U\mathbf{y}\| = s$. The vector $\mathbf{x} = \mathbf{Q}_U\mathbf{y} \in U$ is a unit vector (Lemma 19.8.1 (b)) with $P_U\mathbf{x} = \mathbf{x}$, so $\mathbf{D}\mathbf{x} = (\mathbf{I} - P_W)\mathbf{x}$ and $\|\mathbf{D}\mathbf{x}\| = s$. Hence $\|\mathbf{D}\|_2 = s$.

- (c) By (a) and Proposition 19.8.2 (b), $\|\mathbf{D}\|_2 = \sin \theta_k \leq 1$, with equality if and only if $\cos \theta_k = \sigma_k(\mathbf{M}) = 0$. The square matrix $\mathbf{M}$ has $\sigma_k(\mathbf{M}) = 0$ if and only if $\text{rank } \mathbf{M} < k$ (Theorem 12.8.1), that is, if and only if $\mathbf{M}\mathbf{y} = \mathbf{0}$ for some $\mathbf{y} \neq \mathbf{0}$. Put $\mathbf{u} = \mathbf{Q}_U\mathbf{y}$, which is a non-zero vector of U exactly when $\mathbf{y} \neq \mathbf{0}$, by Lemma 19.8.1 (b). Then $\mathbf{M}\mathbf{y} = \mathbf{Q}_W^*\mathbf{u}$, and $\mathbf{Q}_W^*\mathbf{u} = \mathbf{0}$ if and only if $P_W\mathbf{u} = \mathbf{Q}_W\mathbf{Q}_W^*\mathbf{u} = \mathbf{0}$, again by parts (b) and (a) of the lemma. Finally $P_W\mathbf{u} = \mathbf{0}$ means that $\mathbf{u} \in \ker P_W = W^\perp$, by Theorem 3.10.1 (b) applied to $F^n = W \oplus W^\perp$. So equality holds if and only if U contains a non-zero vector of $W^\perp$. ■

The factor $\sqrt{2}$ in (b) has a clean explanation. When $k = l$, the principal vectors of Proposition 19.8.3 give $P_U = \sum_i \mathbf{u}_i \mathbf{u}_i^*$ and $P_W = \sum_i \mathbf{w}_i \mathbf{w}_i^*$ (Lemma 19.8.1 (a)), so $P_U - P_W$ is a sum of the k terms $\mathbf{u}_i \mathbf{u}_i^* - \mathbf{w}_i \mathbf{w}_i^*$. These act in the mutually orthogonal planes $\text{span}(\mathbf{u}_i, \mathbf{w}_i)$. Each term is Hermitian with trace 0 and rank at most 2, so its non-zero eigenvalues are a pair $\pm t_i$, and being Hermitian it is normal, so $\|\mathbf{u}_i \mathbf{u}_i^* - \mathbf{w}_i \mathbf{w}_i^*\|_F^2 = t_i^2 + t_i^2$ by Theorem 11.11.2. On the other hand, expanding the trace with $\mathbf{u}_i^* \mathbf{u}_i = \mathbf{w}_i^* \mathbf{w}_i = 1$ and $|\langle \mathbf{u}_i, \mathbf{w}_i \rangle| = \cos \theta_i$ (Proposition 19.8.3),

$$\|\mathbf{u}_i \mathbf{u}_i^* - \mathbf{w}_i \mathbf{w}_i^*\|_F^2 = 2 - 2|\langle \mathbf{u}_i, \mathbf{w}_i \rangle|^2 = 2 \sin^2 \theta_i,$$

so $t_i = \sin \theta_i$. Each angle is therefore counted twice in the Frobenius norm and once in the spectral norm. The worked example below shows the $\pm$ pairs explicitly.

Example. (Two Planes in Four Dimensions)

Let $U = \text{span}(\mathbf{e}_1, \mathbf{e}_2) \subseteq \mathbb{R}^4$ and $W = \text{span}(\mathbf{a}, \mathbf{b})$, where

$$\mathbf{a} = (3, 4, 4, 3), \quad \mathbf{b} = (3, -4, 4, -3).$$

Find the principal angles, a pair of principal vectors for each, the sines from Theorem 19.8.4, and $\|P_U - P_W\|$ in both norms. Then check the last two numbers directly.

Solution. Orthonormal bases. $\langle \mathbf{a}, \mathbf{b} \rangle = 9 - 16 + 16 - 9 = 0$ and $\|\mathbf{a}\|^2 = \|\mathbf{b}\|^2 = 50$, so $\mathbf{Q}_W = \frac{1}{5\sqrt{2}}(\mathbf{a} \mid \mathbf{b})$ is an orthonormal basis matrix of W , and $\mathbf{Q}_U = (\mathbf{e}_1 \mid \mathbf{e}_2)$. The matrix $\mathbf{Q}_W^\top \mathbf{Q}_U$ holds the first two entries of $\mathbf{a}$ and $\mathbf{b}$:

$$\mathbf{M} = \mathbf{Q}_W^\top \mathbf{Q}_U = \frac{1}{5\sqrt{2}} \begin{pmatrix} 3 & 4 \\ 3 & -4 \end{pmatrix}, \quad \mathbf{M}^\top \mathbf{M} = \begin{pmatrix} 9/25 & 0 \\ 0 & 16/25 \end{pmatrix}.$$

So the singular values are $4/5$ and $3/5$, and

$$\theta_1 = \arccos \frac{4}{5} \approx 36.87^\circ, \quad \theta_2 = \arccos \frac{3}{5} \approx 53.13^\circ.$$

Note $\sin \theta_1 = 3/5 = \cos \theta_2$, so $\theta_1 + \theta_2 = \pi/2$ here.

Principal vectors. The eigenvector of $\mathbf{M}^\top \mathbf{M}$ for $16/25$ is $\mathbf{z}_1 = \mathbf{e}_2$, and $\mathbf{y}_1 = \mathbf{M}\mathbf{z}_1/\sigma_1 = \frac{1}{5\sqrt{2}}(4, -4) \cdot \frac{5}{4} = \frac{1}{\sqrt{2}}(1, -1)$. By [Proposition 19.8.3](#),

$$\mathbf{u}_1 = \mathbf{e}_2, \quad \mathbf{w}_1 = \mathbf{Q}_W \mathbf{y}_1 = \frac{1}{10}(\mathbf{a} - \mathbf{b}) = (0, \frac{4}{5}, 0, \frac{3}{5}),$$

and $\langle \mathbf{u}_1, \mathbf{w}_1 \rangle = 4/5$. Likewise $\mathbf{z}_2 = \mathbf{e}_1$, $\mathbf{y}_2 = \frac{1}{\sqrt{2}}(1, 1)$, $\mathbf{u}_2 = \mathbf{e}_1$ and $\mathbf{w}_2 = \frac{1}{10}(\mathbf{a} + \mathbf{b}) = (\frac{3}{5}, 0, \frac{4}{5}, 0)$, with $\langle \mathbf{u}_2, \mathbf{w}_2 \rangle = 3/5$, while $\langle \mathbf{u}_1, \mathbf{w}_2 \rangle = \langle \mathbf{u}_2, \mathbf{w}_1 \rangle = 0$. So W is obtained from U by tilting $\mathbf{e}_2$ towards $\mathbf{e}_4$ through θ_1 and $\mathbf{e}_1$ towards $\mathbf{e}_3$ through θ_2 , in two perpendicular planes.

Sines. The vectors $\mathbf{c} = (4, -3, -3, 4)$ and $\mathbf{d} = (4, 3, -3, -4)$ are orthogonal to $\mathbf{a}$, to $\mathbf{b}$ and to each other:

$$\begin{aligned} \langle \mathbf{c}, \mathbf{a} \rangle &= 12 - 12 - 12 + 12 = 0, & \langle \mathbf{c}, \mathbf{b} \rangle &= 12 + 12 - 12 - 12 = 0, \\ \langle \mathbf{d}, \mathbf{a} \rangle &= 12 + 12 - 12 - 12 = 0, & \langle \mathbf{d}, \mathbf{b} \rangle &= 12 - 12 - 12 + 12 = 0, \end{aligned}$$

and $\langle \mathbf{c}, \mathbf{d} \rangle = 16 - 9 + 9 - 16 = 0$. Also $\|\mathbf{c}\|^2 = \|\mathbf{d}\|^2 = 50$. Since $\dim W^\perp = 2$ (Theorem 10.3.2 (c)), $\mathbf{Q}_{W^\perp} = \frac{1}{5\sqrt{2}}(\mathbf{c} \mid \mathbf{d})$ is an orthonormal basis matrix of $W^\perp$, and

$$\mathbf{N} = \mathbf{Q}_{W^\perp}^\top \mathbf{Q}_U = \frac{1}{5\sqrt{2}} \begin{pmatrix} 4 & -3 \\ 4 & 3 \end{pmatrix}, \quad \mathbf{N}^\top \mathbf{N} = \begin{pmatrix} 16/25 & 0 \\ 0 & 9/25 \end{pmatrix}.$$

Its eigenvalues are $16/25 = \sin^2 \theta_2$ and $9/25 = \sin^2 \theta_1$, and $\mathbf{M}^\top \mathbf{M} + \mathbf{N}^\top \mathbf{N} = \mathbf{I}_2$, as [Theorem 19.8.4](#) (a) says.

Distances. By [Theorem 19.8.5](#), $\|P_U - P_W\|_2 = \sin \theta_2 = 4/5$ and $\|P_U - P_W\|_F = \sqrt{2}(\frac{9}{25} + \frac{16}{25})^{1/2} = \sqrt{2}$.

Direct check. By [Lemma 19.8.1](#) (a), $P_W = \mathbf{Q}_W \mathbf{Q}_W^\top = \frac{1}{50}(\mathbf{a}\mathbf{a}^\top + \mathbf{b}\mathbf{b}^\top)$, and

$$P_W = \frac{1}{50} \begin{pmatrix} 18 & 0 & 24 & 0 \\ 0 & 32 & 0 & 24 \\ 24 & 0 & 32 & 0 \\ 0 & 24 & 0 & 18 \end{pmatrix}, \quad P_U = \text{diag}(1, 1, 0, 0).$$

In $\mathbf{D} = P_U - P_W$, coordinates 1, 3 do not interact with coordinates 2, 4, and the two 2×2 blocks are

$$\frac{1}{50} \begin{pmatrix} 32 & -24 \\ -24 & -32 \end{pmatrix}, \quad \frac{1}{50} \begin{pmatrix} 18 & -24 \\ -24 & -18 \end{pmatrix}.$$

Each is symmetric with trace 0, so its eigenvalues are $\pm\sqrt{-\det}$: $\pm\frac{1}{50}\sqrt{32^2 + 24^2} = \pm\frac{4}{5}$ and $\pm\frac{1}{50}\sqrt{18^2 + 24^2} = \pm\frac{3}{5}$. So $\mathbf{D}$ has eigenvalues $\pm\frac{4}{5}, \pm\frac{3}{5}$. By [Lemma 16.3.5](#), $\|\mathbf{D}\|_2 = 4/5$, and $\|\mathbf{D}\|_F^2 = \text{tr}(\mathbf{D}^2) = 2(\frac{16}{25} + \frac{9}{25}) = 2$. Both agree with the theorem, and the eigenvalues come in the $\pm \sin \theta_i$ pairs described before the example.

Remark. A metric, and why the dimensions must match. On the set of k -dimensional subspaces of F^n , the number $d(U, W) = \|P_U - P_W\|_2 = \sin \theta_k$ is a metric. It is symmetric. It is zero only when $P_U = P_W$, that is, when $U = \text{im } P_U = \text{im } P_W = W$. And it satisfies the triangle inequality because $\|\cdot\|_2$ is a norm ([Theorem 15.3.2](#) (b)). Between subspaces of **different** dimensions it is useless. If $k < l$, then $\dim(W \cap U^\perp) \geq l + (n - k) - n > 0$ by [Theorem 1.7.3](#), so some unit $\mathbf{w} \in W$ is perpendicular to U . For it

$(P_U - P_W)\mathbf{w} = -\mathbf{w}$, so $\|P_U - P_W\|_2 \geq 1$. On the other hand, [Eq. 19.8.1](#), which did not use the dimensions, together with $\|P\mathbf{v}\| \leq \|\mathbf{v}\|$ for orthogonal projections (Theorem 10.3.3 (c)), gives $\|P_U - P_W\|_2 \leq 1$. So the distance is exactly 1, whatever the principal angles are. This is why Section 9 always compares invariant subspaces of the same dimension.

19.8.5 Exercises

19.8.5.1 A. Check your understanding

Exercise A1.

- Define the principal angles between subspaces $U, W \subseteq F^n$ with $1 \leq \dim U \leq \dim W$.
- True or false: if $U \subseteq W$, then every principal angle between U and W is 0. Justify your answer.
- True or false: the principal angle between $\text{span}(\mathbf{u})$ and $\text{span}(-\mathbf{u})$, for $\mathbf{u} \neq \mathbf{0}$, is π . Justify your answer.
- True or false: if $U \cap W \neq \{\mathbf{0}\}$, then $\theta_1 = 0$. Justify your answer.
- State the formula for $\|P_U - P_W\|_2$ in terms of principal angles, with its hypothesis.
- True or false: for two distinct lines U, W in $\mathbb{R}^2$, $\|P_U - P_W\|_2 < 1$. Justify your answer.

Solution.

- Choose orthonormal basis matrices $\mathbf{Q}_U \in M_{n \times k}(F)$ and $\mathbf{Q}_W \in M_{n \times l}(F)$. The principal angles $0 \leq \theta_1 \leq \dots \leq \theta_k \leq \pi/2$ are defined by $\cos \theta_i = \sigma_i(\mathbf{Q}_W^* \mathbf{Q}_U)$, $i = 1, \dots, k$ ([Definition 19.8.1](#)). They do not depend on the bases ([Proposition 19.8.2](#)).
- True. As in Example 2 after [Proposition 19.8.2](#), $\mathbf{M}^* \mathbf{M} = \mathbf{Q}_U^* P_W \mathbf{Q}_U = \mathbf{Q}_U^* \mathbf{Q}_U = \mathbf{I}_k$, so every $\cos \theta_i = 1$.
- False. The two lines are equal, so the angle is 0: with $\mathbf{u}_0 = \mathbf{u}/\|\mathbf{u}\|$, $|\langle \mathbf{u}_0, -\mathbf{u}_0 \rangle| = 1$ ([Proposition 19.8.2](#) (d)). The value π is the angle between the vectors $\mathbf{u}$ and $-\mathbf{u}$ of Definition 10.1.5, which is a different notion.
- True. Take a unit $\mathbf{u} \in U \cap W$ and write $\mathbf{u} = \mathbf{Q}_U \mathbf{y}$ with $\|\mathbf{y}\| = 1$ ([Lemma 19.8.1](#) (b)). Then $\|\mathbf{M} \mathbf{y}\| = \|\mathbf{Q}_W^* \mathbf{u}\| = \|P_W \mathbf{u}\| = \|\mathbf{u}\| = 1$, so $\sigma_1(\mathbf{M}) = \|\mathbf{M}\|_2 \geq 1$. With [Proposition 19.8.2](#) (b), $\cos \theta_1 = 1$.
- If $\dim U = \dim W = k \geq 1$, then $\|P_U - P_W\|_2 = \sin \theta_k$, the sine of the largest principal angle ([Theorem 19.8.5](#) (a)).
- False. The two axes $U = \text{span}(\mathbf{e}_1)$ and $W = \text{span}(\mathbf{e}_2)$ have $U \cap W^\perp = U \neq \{\mathbf{0}\}$, so $\|P_U - P_W\|_2 = 1$ by [Theorem 19.8.5](#) (c). Directly, $P_U - P_W = \text{diag}(1, -1)$.

19.8.5.2 B. Practice

Exercise B1: Two planes in three dimensions.

Let $U = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ and $W = \text{span}((1, 0, 0), (1, 3, 4))$ in $\mathbb{R}^3$.

- Find an orthonormal basis matrix of W .

- (b) Compute the principal angles between U and W .
- (c) Hence compute $\|P_U - P_W\|_2$ and $\|P_U - P_W\|_F$.
- (d) Explain why $\theta_1 = 0$ could have been predicted without computation.

Solution.

- (a) Gram–Schmidt: $\mathbf{q}_1 = \mathbf{e}_1$, and $(1, 3, 4) - \langle (1, 3, 4), \mathbf{e}_1 \rangle \mathbf{e}_1 = (0, 3, 4)$, of length 5, so $\mathbf{q}_2 = (0, \frac{3}{5}, \frac{4}{5})$ and $\mathbf{Q}_W = (\mathbf{q}_1 \mid \mathbf{q}_2)$.
- (b) With $\mathbf{Q}_U = (\mathbf{e}_1 \mid \mathbf{e}_2)$, the matrix $\mathbf{M} = \mathbf{Q}_W^\top \mathbf{Q}_U$ holds the first two entries of $\mathbf{q}_1$ and $\mathbf{q}_2$:

$$\mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & 3/5 \end{pmatrix}.$$

It is diagonal with non-negative entries, so $\mathbf{M}^\top \mathbf{M} = \text{diag}(1, \frac{9}{25})$ and the singular values are 1 and $\frac{3}{5}$. So $\theta_1 = 0$ and $\theta_2 = \arccos \frac{3}{5} \approx 53.13^\circ$, with $\sin \theta_2 = \frac{4}{5}$.

- (c) By [Theorem 19.8.5](#), $\|P_U - P_W\|_2 = \sin \theta_2 = \frac{4}{5}$ and $\|P_U - P_W\|_F = \sqrt{2} (0 + \frac{16}{25})^{1/2} = \frac{4\sqrt{2}}{5}$.
- (d) Two planes in $\mathbb{R}^3$ meet in a line (Theorem 1.7.3, as in the opening of the section), and a common unit vector forces $\cos \theta_1 = 1$ (Exercise A1 (d)). Here the common line is $\text{span}(\mathbf{e}_1)$.

Exercise B2: Which projections are at distance one?.

Determine, for each pair, whether $\|P_U - P_W\|_2 = 1$. Justify your answer.

- (a) $U = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ and $W = \text{span}(\mathbf{e}_2, \mathbf{e}_3)$ in $\mathbb{R}^3$.
- (b) $U = \text{span}((1, 1, 0))$ and $W = \text{span}((1, -1, 1))$ in $\mathbb{R}^3$.
- (c) $U = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ and $W = \text{span}((1, 0, 1), (0, 1, 1))$ in $\mathbb{R}^3$.
- (d) $U = \text{span}((1, 2, 3))$ and $W = \text{span}((1, 0, 0), (0, 1, 0))$ in $\mathbb{R}^3$.

Solution.

- (a) Yes. $W^\perp = \text{span}(\mathbf{e}_1) \subseteq U$, so $U \cap W^\perp \neq \{\mathbf{0}\}$, and [Theorem 19.8.5](#) (c) applies since both dimensions are 2.
- (b) Yes. $\langle (1, 1, 0), (1, -1, 1) \rangle = 0$, so $U \subseteq W^\perp$, and (c) of the theorem applies with $k = 1$.
- (c) No. Both are planes, and $W^\perp = \text{span}((1, 1, -1))$, since $(1, 1, -1)$ is orthogonal to both spanning vectors of W and $\dim W^\perp = 1$. This vector has non-zero third entry, so it is not in U , and $U \cap W^\perp = \{\mathbf{0}\}$. By the theorem, $\|P_U - P_W\|_2 < 1$. (In fact the principal angles are 0 and $\arccos(1/\sqrt{3})$, so the distance is $\sqrt{2/3}$.)
- (d) Yes, for a different reason: the dimensions are 1 and 2, and the remark after [Example \(Two Planes in Four Dimensions\)](#) shows that $\|P_U - P_W\|_2 = 1$ whenever the dimensions differ. The theorem itself does not apply. Its principal angle $\theta_1 = \arccos(\sqrt{5}/\sqrt{14})$ is far from $\pi/2$, which illustrates why the projection distance is only used for equal dimensions.

Exercise B3: Two lines.

Let $\mathbf{u}, \mathbf{w} \in F^n$ be unit vectors and $c = |\langle \mathbf{u}, \mathbf{w} \rangle|$. Prove that

$$\|\mathbf{u}\mathbf{u}^* - \mathbf{w}\mathbf{w}^*\|_2 = \sqrt{1 - c^2}, \quad \|\mathbf{u}\mathbf{u}^* - \mathbf{w}\mathbf{w}^*\|_F = \sqrt{2} \sqrt{1 - c^2}.$$

Solution. Let $U = \text{span}(\mathbf{u})$ and $W = \text{span}(\mathbf{w})$, both of dimension 1. By [Lemma 19.8.1](#) (a) with $\mathbf{Q}_U = \mathbf{u}$ and $\mathbf{Q}_W = \mathbf{w}$, $P_U = \mathbf{u}\mathbf{u}^*$ and $P_W = \mathbf{w}\mathbf{w}^*$. By [Proposition 19.8.2](#) (d), the single principal angle satisfies $\cos \theta_1 = c$, so $\sin \theta_1 = \sqrt{1 - c^2}$, since $\sin \theta_1 \geq 0$ on $[0, \pi/2]$. Now [Theorem 19.8.5](#) (a) and (b) with $k = 1$ give $\|P_U - P_W\|_2 = \sin \theta_1$ and $\|P_U - P_W\|_F = \sqrt{2} \sin \theta_1$, which are the two claims.

19.8.5.3 C. Going deeper**Exercise C1: Counting zero and right angles.**

Let $U, W \subseteq F^n$ with $1 \leq k = \dim U \leq l = \dim W$, orthonormal basis matrices $\mathbf{Q}_U, \mathbf{Q}_W$, and $\mathbf{M} = \mathbf{Q}_W^* \mathbf{Q}_U$.

- (a) Prove that $\mathbf{y}^*(\mathbf{I}_k - \mathbf{M}^* \mathbf{M})\mathbf{y} = \|(\mathbf{I} - P_W)\mathbf{Q}_U \mathbf{y}\|^2$ for every $\mathbf{y} \in F^k$.
- (b) Deduce that the number of principal angles equal to 0 is $\dim(U \cap W)$.
- (c) Prove that the number of principal angles equal to $\pi/2$ is $\dim(U \cap W^\perp)$.

Hint for (b): a positive semidefinite matrix $\mathbf{B}$ with $\mathbf{y}^ \mathbf{B} \mathbf{y} = 0$ has $\mathbf{B} \mathbf{y} = \mathbf{0}$.*

Solution.

- (a) Let $\mathbf{u} = \mathbf{Q}_U \mathbf{y}$. By [Lemma 19.8.1](#) (a) and (b), $\mathbf{y}^* \mathbf{y} = \|\mathbf{u}\|^2$ and $\mathbf{y}^* \mathbf{M}^* \mathbf{M} \mathbf{y} = \|\mathbf{Q}_W^* \mathbf{u}\|^2 = \|P_W \mathbf{u}\|^2$. Since $\mathbf{u} = P_W \mathbf{u} + (\mathbf{I} - P_W)\mathbf{u}$ with the two parts orthogonal (Theorem 10.3.2), Theorem 10.1.3 gives $\|\mathbf{u}\|^2 - \|P_W \mathbf{u}\|^2 = \|(\mathbf{I} - P_W)\mathbf{u}\|^2$.
- (b) Put $\mathbf{B} = \mathbf{I}_k - \mathbf{M}^* \mathbf{M}$. It is Hermitian, and by (a) it is positive semidefinite (Definition 12.1.1). Its eigenvalues are $1 - \cos^2 \theta_i = \sin^2 \theta_i$, with an orthonormal eigenbasis $\mathbf{z}_1, \dots, \mathbf{z}_k$ (as in the proof of [Theorem 19.8.4](#)). So the number of i with $\theta_i = 0$ is the number of zero eigenvalues, which is $\dim \ker \mathbf{B}$.
We claim $\ker \mathbf{B} = \{\mathbf{y} : \mathbf{Q}_U \mathbf{y} \in W\}$. If $\mathbf{B} \mathbf{y} = \mathbf{0}$, then $\mathbf{y}^* \mathbf{B} \mathbf{y} = 0$, so $(\mathbf{I} - P_W)\mathbf{Q}_U \mathbf{y} = \mathbf{0}$ by (a), and $\mathbf{Q}_U \mathbf{y} = P_W \mathbf{Q}_U \mathbf{y} \in W$. Conversely, if $\mathbf{Q}_U \mathbf{y} \in W$, then $\mathbf{y}^* \mathbf{B} \mathbf{y} = 0$ by (a). Writing $\mathbf{y} = \sum_i c_i \mathbf{z}_i$, this says $\sum_i \sin^2 \theta_i |c_i|^2 = 0$. So $c_i = 0$ whenever $\sin \theta_i \neq 0$, and then $\mathbf{B} \mathbf{y} = \sum_i c_i \sin^2 \theta_i \mathbf{z}_i = \mathbf{0}$. This proves the claim; the second half of it is the fact in the hint, proved here for this $\mathbf{B}$.
The map $\mathbf{y} \mapsto \mathbf{Q}_U \mathbf{y}$ is linear and injective ([Lemma 19.8.1](#) (b)). It carries $\ker \mathbf{B}$ onto $\{\mathbf{u} \in U : \mathbf{u} \in W\} = U \cap W$, because every vector of U is $\mathbf{Q}_U \mathbf{y}$ for some $\mathbf{y}$. So $\dim \ker \mathbf{B} = \dim(U \cap W)$.
- (c) $\theta_i = \pi/2$ exactly when $\sigma_i(\mathbf{M}) = 0$. The number of non-zero singular values of $\mathbf{M}$ is $\text{rank } \mathbf{M}$ (Theorem 12.8.1), so the number of zero ones among the k is $k - \text{rank } \mathbf{M} = \dim \ker \mathbf{M}$, by the Rank–Nullity Theorem (Theorem 3.3.1) for $\mathbf{y} \mapsto \mathbf{M} \mathbf{y}$ on F^k . Next, $\mathbf{M} \mathbf{y} = \mathbf{Q}_W^* \mathbf{Q}_U \mathbf{y} = \mathbf{0}$ if and only if $P_W \mathbf{Q}_U \mathbf{y} = \mathbf{Q}_W (\mathbf{Q}_W^* \mathbf{Q}_U \mathbf{y}) = \mathbf{0}$, because $\mathbf{Q}_W$ is injective. That holds if and only if $\mathbf{Q}_U \mathbf{y} \in W^\perp$. As in (b), $\mathbf{y} \mapsto \mathbf{Q}_U \mathbf{y}$ carries $\ker \mathbf{M}$ injectively onto $U \cap W^\perp$, so the count is $\dim(U \cap W^\perp)$.

Exercise C2: The smallest angle as a maximum.

Let $U, W \subseteq F^n$ with $1 \leq k = \dim U \leq l = \dim W$ and principal angles $\theta_1 \leq \dots \leq \theta_k$.

- (a) Prove that

$$\cos \theta_1 = \max\{|\langle \mathbf{u}, \mathbf{w} \rangle| : \mathbf{u} \in U, \mathbf{w} \in W, \|\mathbf{u}\| = \|\mathbf{w}\| = 1\},$$

and that the maximum is attained at the principal vectors $\mathbf{u}_1, \mathbf{w}_1$.

- (b) Deduce that θ_1 is the smallest principal angle between a line of U and a line of W , which is the notion tried and rejected at the start of the section.
- (c) Hence explain, in one sentence, why θ_1 alone cannot distinguish the pairs (U, W_1) and (U, U) of the opening, but the full list θ_1, θ_2 can.

Solution.

- (a) Let $\mathbf{u} \in U$ and $\mathbf{w} \in W$ be unit vectors. By [Lemma 19.8.1](#) (a) and (b), $\mathbf{u} = \mathbf{Q}_U \mathbf{y}$ and $\mathbf{w} = \mathbf{Q}_W \mathbf{z}$ with $\mathbf{y} = \mathbf{Q}_U^* \mathbf{u}$ and $\mathbf{z} = \mathbf{Q}_W^* \mathbf{w}$ unit vectors. Then $\langle \mathbf{u}, \mathbf{w} \rangle = \mathbf{z}^* \mathbf{Q}_W^* \mathbf{Q}_U \mathbf{y} = \mathbf{z}^* \mathbf{M} \mathbf{y}$, and by Cauchy–Schwarz and Theorem 15.3.2 (a),

$$|\langle \mathbf{u}, \mathbf{w} \rangle| \leq \|\mathbf{z}\| \|\mathbf{M} \mathbf{y}\| \leq \|\mathbf{M}\|_2 = \sigma_1(\mathbf{M}) = \cos \theta_1,$$

using Theorem 15.3.3 (c). By [Proposition 19.8.3](#), the unit vectors $\mathbf{u}_1 \in U$ and $\mathbf{w}_1 \in W$ satisfy $\langle \mathbf{u}_1, \mathbf{w}_1 \rangle = \cos \theta_1$, so the maximum exists and equals $\cos \theta_1$.

- (b) By [Proposition 19.8.2](#) (d), the principal angle between the lines $\text{span}(\mathbf{u})$ and $\text{span}(\mathbf{w})$ is $\arccos|\langle \mathbf{u}, \mathbf{w} \rangle|$. Since $\arccos$ is decreasing, the smallest such angle corresponds to the largest $|\langle \mathbf{u}, \mathbf{w} \rangle|$, which by (a) is $\cos \theta_1$.
- (c) Both pairs share a line, so both have $\theta_1 = 0$, and the first number cannot separate them. The second angle is $\pi/2$ for (U, W_1) and 0 for (U, U) , by Examples 4 and 2 after [Proposition 19.8.2](#).